

Selectime: validation-selected input and retrieval mixtures

30 September 2026

Evidence status

The completed comparison contains 90 matched TIME tasks and 67,502 Seasonal-defined metric rows per method. The numerical tables and figures in this report describe the completed artifact snapshot published as Git revision 924bece57. They are marked [†] because two subsequently implemented calibration gates—minimum validation support and minimum empirical alternative win rate—have not yet been applied to the saved results. Candidate forecasts remain reusable, but calibration, assembly, evaluation, and reporting must be regenerated before these values become the current gated result.

Main findings from the completed pre-gate snapshot

Chronos-2 scope mixing was strongest overall. The per-variate scope mixture reduced mean task MASE by **1.013%** relative to univariate input, while the task scope mixture had the best task win rate at 80.0%. Native multivariate input remained competitive at **0.958%**. Among retrieval mixtures, $K = 5$ was best in aggregate at **0.363%**; its advantage was concentrated on medium and long horizons. No single K dominated every frequency–horizon cell.

Chronos-Bolt reported only vanilla univariate forecasting because its backbone does not accept non-empty covariates or multivariate histories. Its mean task MASE was 1.167905, 9.23% above univariate Chronos-2. This is a backbone comparison, not evidence that the Selectime controls were tested on Bolt.

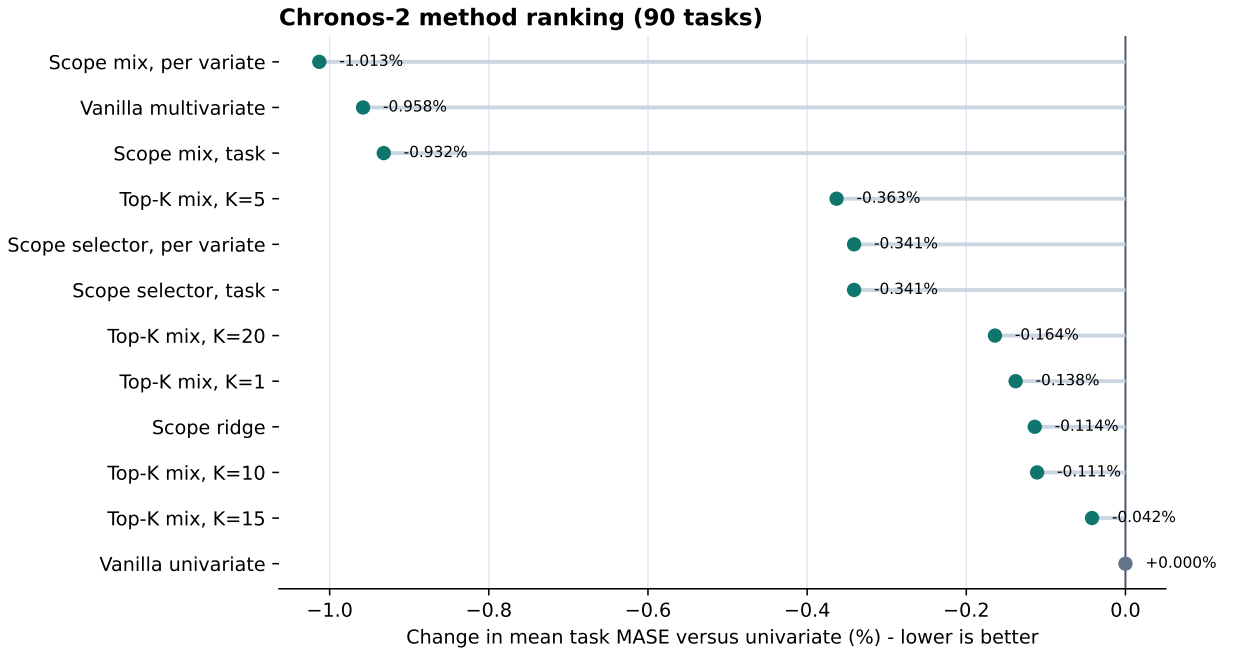


Figure 1: Completed pre-gate Chronos-2 ranking. Negative change means lower mean task MASE than univariate Chronos-2. Every method gives equal weight to each of the 90 task means.

1 Question and reduced protocol

For target history x and horizon- H future y , let V be the frozen univariate forecast and M the native multivariate Chronos-2 forecast. For $K \in \{1, 5, 10, 15, 20\}$, retrieve the K closest aligned historical windows by instance-normalized lookback distance. Rescale their histories and already observed futures using lookback-only statistics, then supply them as past and future covariate channels to obtain R_K . A reported retrieval mixture is

$$\hat{y}_K = (1 - p_K)V + p_K R_K.$$

Scope mixtures replace R_K with M and fit one task-level or item/variate weight. Hard scope selectors choose V or M ; scope ridge fits $V + p(M - V)$ with fixed $\alpha = 1$ and does not clip p .

In the completed snapshot, a heuristic mixture with w validation wins in n eligible validation rows used the Beta(1, 1) posterior mean $p = (1 + w)/(2 + n)$; ties contributed half a win. A task-level weight uses only that task. A per-variate weight uses only the corresponding item/variant. Rows where the alternative fell back to univariate were excluded rather than counted as ties. The canonical univariate forecast has no fallback path and must be finite.

The current unexecuted contract additionally requires

$$n \geq 10, \quad n/N_{\text{test}} > 0.10, \quad w/n > 0.10.$$

Otherwise $p = 0$ and the reported forecast is exactly V . Here N_{test} counts matching forecast rows at the same task or item/variant granularity.

2 Exact leading results

Rank	Method	Mean MASE	Improvement	Win rate	Recorded inference
1	Scope mix, per variate [†]	1.058395	+1.013%	71.1%	Assembled; no independent latency
2	Vanilla multivariate	1.058981	+0.958%	57.8%	87.1 s summed query-loop time
3	Scope mix, task [†]	1.059264	+0.932%	80.0%	Assembled; no independent latency
4	Top- K mix, $K = 5$ [†]	1.065347	+0.363%	56.7%	Assembled; no independent latency
5	Scope selector, per variate	1.065584	+0.341%	41.1%	Assembled; no independent latency

Improvement is $100(1 - \text{MASE}_m/\text{MASE}_V)$, so positive is better. Win rate is the strict fraction of the 90 task MASE values below the univariate task value; ties remain in the denominator. Assembled methods reuse candidate artifacts and therefore have no measured cold inference latency.

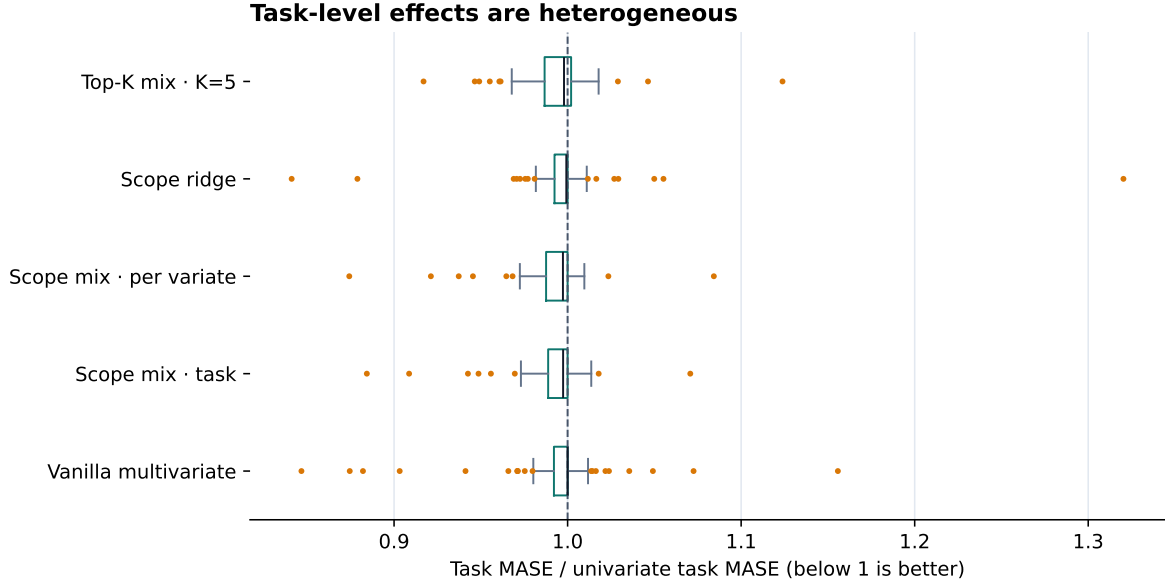


Figure 2: Paired task ratios reveal heterogeneous effects hidden by aggregate means. Values below one improve on univariate; the boxes summarize 90 task ratios and orange points are Tukey outliers.

Largest per-variate-mix gains	Improvement	Largest losses	Degradation
Coastal_T_S / 20T / long	+12.58%	US_Labor / M / short	8.43%
Vehicle_Supply / M / short	+7.88%	Uncertainty_1M / M / short	2.35%
JOLTS / M / short	+6.27%	EWELD_Load / 15T / long	0.96%
Global_Price / Q / short	+5.46%	EWELD_Load / 15T / medium	0.62%
Housing_Inventory / M / short	+3.53%	Coastal_T_S / 20T / short	0.43%

3 Forecast-term and domain structure

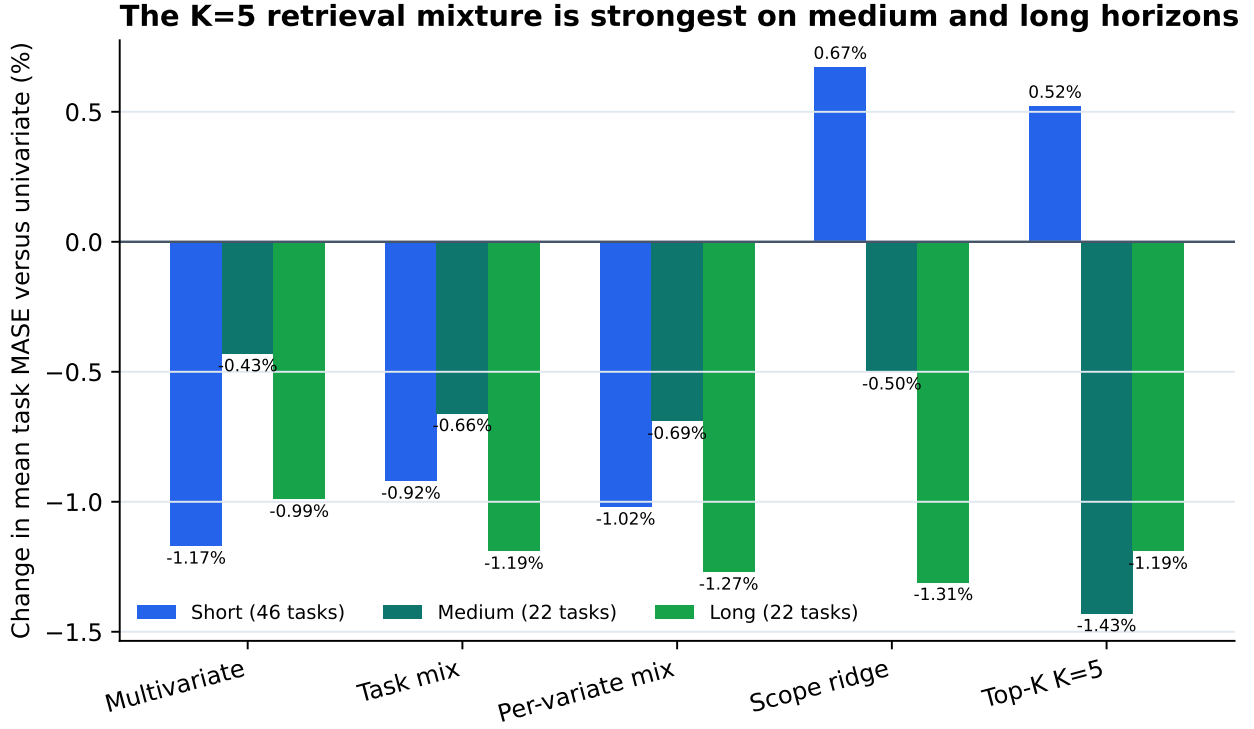


Figure 3: Completed pre-gate changes by forecast term. The $K = 5$ retrieval mixture degrades the short-term mean by 0.52%, but improves medium and long means by 1.43% and 1.19%.

Domain	Tasks	Multivariate	Per-variate mix [†]	$K = 5$ mix [†]
Climate	45	+0.84%	+1.12%	+1.09%
Economics	11	+2.79%	+2.16%	-0.89%
Transport	11	-0.14%	+0.05%	+0.26%
Energy	6	-1.02%	+0.26%	+2.14%
Finance	6	+1.64%	+0.88%	-0.22%
Sales	4	-0.08%	+0.27%	-0.28%
Healthcare	3	+0.20%	+0.46%	-1.71%
Industry	3	+0.02%	+0.02%	0.00%
Cloud operations	1	-0.55%	-0.02%	0.00%

Positive values in this table mean improvement over univariate. Small-domain rows are descriptive; they are not separately powered comparisons.

4 Frequency–horizon winners

5 Retrieval mixtures and fallback

K	Mean MASE	Improvement	Win/loss/tie	Same-series neighbors	Norm. calendar lag
1	1.067750	+0.138%	48/32/10	46.3%	0.450
5	1.065347	+0.363%	51/29/10	43.9%	0.459
10	1.068043	+0.111%	48/32/10	41.9%	0.469
15	1.068782	+0.042%	50/30/10	40.3%	0.475
20	1.067474	+0.164%	49/31/10	38.9%	0.479

All five mixtures had a 4.612% active retrieval fallback rate. For $K = 5$, the raw candidate lacked enough neighbors on 10,293 rows. Ten tasks already had $p = 0$ because they had no usable validation rows, so only 3,113 unavailable test rows had a scientifically active nonzero retrieval weight and counted as assembled-method fallback.

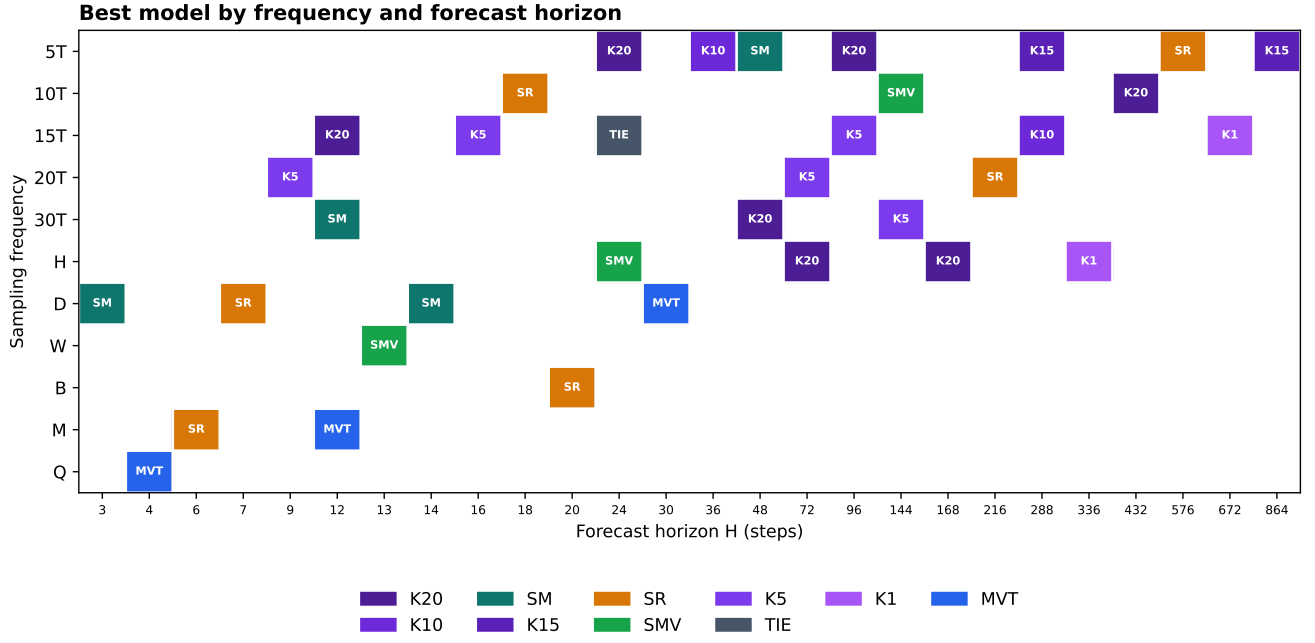


Figure 4: Best completed pre-gate method at each sampling-frequency and forecast-horizon cell, using the same tasks inside each cell. Codes: MVT native multivariate; SM/SMV task/per-variate scope mix; SR scope ridge; Kx retrieval mixture; TIE exact tie between univariate and scope selector. Sparse cells may represent only one task.

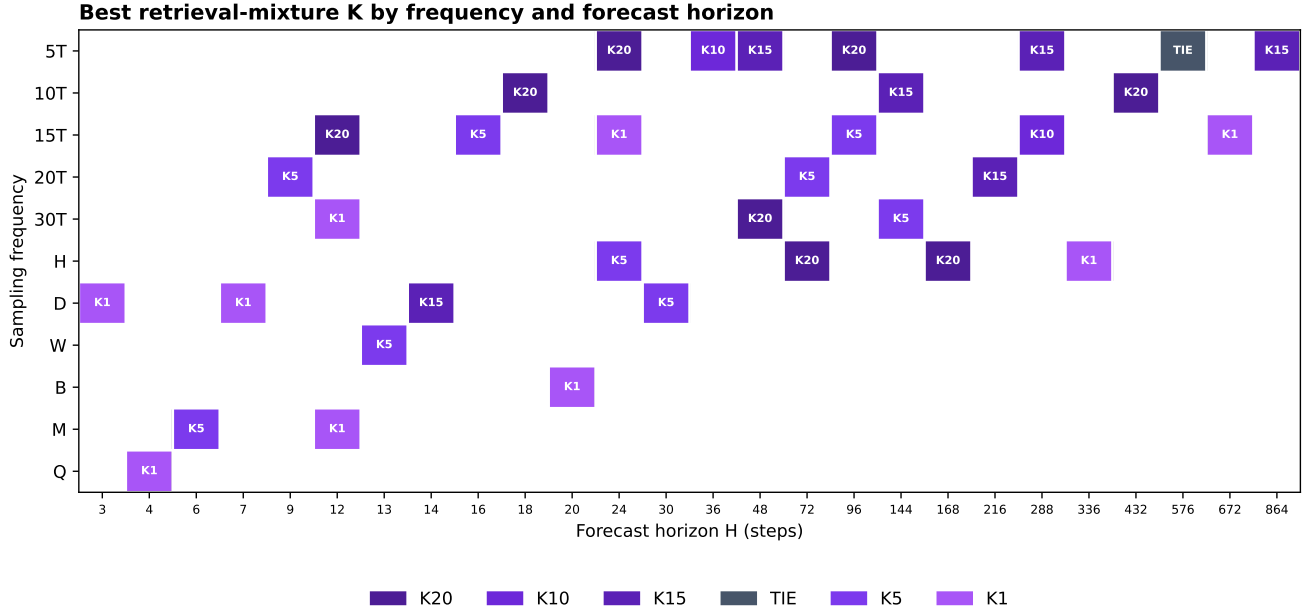


Figure 5: Best K when comparing only the five retrieval mixtures. Across 35 cells, $K = 1$ and $K = 5$ each win nine, $K = 20$ wins eight, $K = 15$ wins six, $K = 10$ wins two, and one cell ties. Aggregate $K = 5$ is therefore a practical default, not a universal optimum.

Mixture state	Intended forecast	Unavailable alternative	Fallback count
$p = 0$	V	Alternative is unused	No: output was already exactly V
$p \neq 0$	$(1 - p)V + pR$	Emit V	Yes: the intended mixture could not be formed

6 Calibration distribution and new safeguards

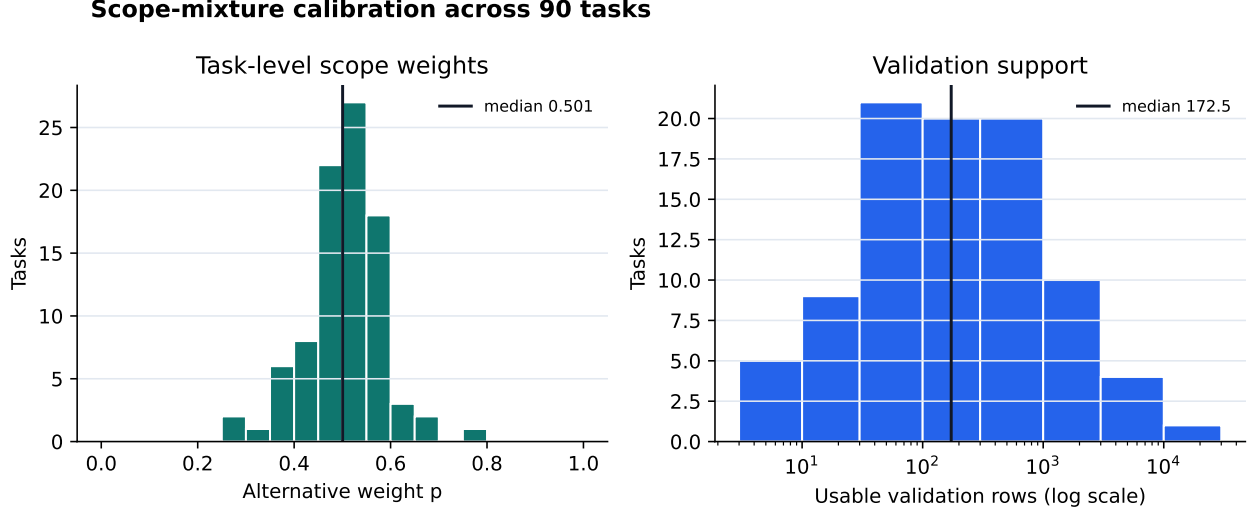


Figure 6: Completed pre-gate task-scope calibration. The weight median is 0.501 (IQR 0.470–0.554; range 0.281–0.787). Usable validation rows have median 172.5 (IQR 41.25–519.75; range 4–14,868). The hard task selector separately chose multivariate on 29 tasks and univariate on 61.

Beta smoothing alone assigns nonzero weight after zero observed wins. The new gates make that prior contribution ineligible and also reject poorly supported calibrations:

Usable n	Wins w	Ungated Beta p	Current p	Reason
1	0	33.33%	0	$n < 10$ and no wins
5	0	14.29%	0	$n < 10$ and no wins
10	0	8.33%	0	$w/n \leq 10\%$
20	0	4.55%	0	$w/n \leq 10\%$
50	0	1.92%	0	$w/n \leq 10\%$

The relative-support gate additionally requires $n/N_{\text{test}} > 10\%$. At task level, the completed snapshot had no empirical win rate at or below 10%; the support gate can still change low- n tasks. At per-variate level, 240 of 3,882 completed selections had positive support but empirical win rate at or below 10%, so the new global gate is not result-neutral.

7 Why the earlier convex and direct-horizon methods differ

Property	Archived Adaptation convex fit	Selectime scope mix
Forecast	$\lambda_0 V + \sum_m \lambda_m Z_m$	$(1 - p)V + pM$
Parameters	$p + 1$ simplex weights; eight for full $K = 3$	One scalar per task or item/variant
Fit target	Constrained least squares on pooled T1+T2	Validation-window MSSE win frequency
Regularization	Simplex only; no ridge penalty	Beta smoothing plus current eligibility gates
Observed evidence	−0.05% to +0.88%; below matched ridge	Pre-gate +0.932% task and +1.013% per-variate
Interpretation	More parameters may overfit, but candidates, splits, losses and datasets also differ	Simpler probability-like weight; not an MSSE-optimal coefficient

The older Selectime `top1_horizon_mix` was different again: it blended V directly with the rescaled observed continuation of the single nearest neighbor and bypassed the backbone. One task produced a six-order-of- magnitude

failure; even after removing that task, the method did not beat vanilla. The current R_K instead results from in-model covariate conditioning on K retrieved examples before the outer mixture.

8 Chronos-Bolt and timing boundary

Bolt method	Tasks	Mean MASE	Scaled MASE	Wins vs C2 univariate
Vanilla univariate	90	1.167905	0.776717	8/90

Chronos-Bolt cannot produce R_K : its adapter rejects covariates and multivariate histories. Treating neighbors as a batch would still forecast them independently and would not condition the query on them. A post-hoc neighbor-horizon mixture would be a new model-agnostic method, not the current Selectime top- K method.

The archived Adaptime timing path was designed to launch a fresh process for each method and example, without cached predictions, representations or neighbors. It was never run to a synchronized result. Current selected-method latencies therefore remain undefined; recorded candidate query-loop times and artifact-reuse overhead are not cold inference benchmarks.

9 Rerun and future experiments

Rerun case	Estimated compute time	Evidence and boundary
Chronos-2 gate-only re-fresh	15–30 minutes, excluding queue	Reuse validation/test candidate forecasts; rerun calibration, assembly, evaluation and report. Previous calibration alone took about 90 seconds.
Cold Chronos-2 full workflow	6–8 hours	The completed launch spent about 3 h 45 min in remaining validation prediction and was still in remaining test prediction more than 5 h 45 min after start.
Chronos-Bolt	No gate-driven rerun	It has no mixture calibration.
TS-ICL	Not yet estimable reliably	Depends on the terminal reusable candidate coverage of the current recovery.

Three follow-up studies are scientifically motivated but have not been run:

1. **Gate refresh.** Regenerate the reduced Selectime comparison under the n , support-ratio and empirical-win gates before promoting any pre-gate number to the current result.
2. **One-axis retrieval ablations on the best retrieval model.** Fix Chronos-2 `top_k_5_mix`, then vary one TS-RAG-like choice at a time: all-series versus same-series scope, aligned versus unrestricted dates, instance- L_2 versus the alternative representation/distance, and query-scale rescaling versus its removal. This separates which retrieval choice matters while keeping $K = 5$ fixed.
3. **Test-fit oracles.** Fit the task/per-variate selector, mixture weight, and ridge coefficient directly on the test losses. These are not deployable forecasts: they intentionally leak test labels. Their value is an optimistic upper bound on the improvement available from perfect selection or calibration among the already computed candidates. Report both the oracle score and the gap from validation-fitted performance.

10 Limits

The completed comparison uses one deterministic seed and provides no across-run uncertainty or significance test. Frequency–horizon cells and domains have unequal and sometimes very small task counts. The new gates are implemented but unevaluated, so starred mixture values are not current gated results. The proposed test-fit oracles would be diagnostic upper bounds only, not honest model estimates. These limits preserve the descriptive value of the completed snapshot while defining the exact rerun needed next.